Quiz for Week 1

Submit your assignment

Due Feb 18, 11:59 PM IST

Try again

Receive grade

To Pass 80% or higher

Your grade 100%

View Feedback

We keep your highest score

Congratulations! You passed!

Grade received 100%

Latest Submission Grade 100%

To pass 80% or higher

Go to next item

1. (**Special note:** Below there are five questions for you to answer. Naturally there are many mathematical formulas and equations in the problem statements. However, due to some technical issues on Coursera, from time to time for some learners, some formulas and equations cannot be displayed correctly (while some others can). As there is no way for the instructing team to solve this issue, we post all the problem statements in a PDF file here. If unfortunately some formulas and equations are not readable for you, please use the PDF file to prepare your answers and then come back to Coursera to choose/fill in your answers.) **1 / 1 point**

Among the following terms, which is not going to be introduced in this course?

- ○ Advanced solvers.
- ○ Exact algorithms.
- ◉ Data mining.
- ○ Heuristic algorithms.
- ○ None of the above.

Correct

2. Among the following terms, which is not a characteristic of heuristic algorithms? **1 / 1 point**

- ○ Completing execution in a reasonably short time.
- ◉ Finding an optimal solution every time.
- ○ Finding a feasible solution when the problem is too complicated.
- ○ All of the above.
- ○ None of the above.

Correct

3. Please solve the following linear system by Gaussian elimination **1 / 1 point**

$$\begin{aligned} u + 2v + 3w &= 19 \\ 4u + 3v + 2w &= 41 \\ 2u - 6v + w &= -2. \end{aligned}$$

What are the values of u, v, and w?

- ○ $u = 7$, $v = 3$, $w = 1$.
- ○ $u = 4$, $v = 2$, $w = 1$.
- ○ $u = 7$, $v = 2$, $w = 2$.
- ◉ $u = 7$, $v = 3$, $w = 2$.
- ○ The system is singular.

Correct

4. Regarding the following three vectors, which statement is not true? **1 / 1 point**

$$x = \begin{bmatrix} 2 \\ 4 \\ -2 \end{bmatrix}, y = \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix}, z = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$$

- ○ If we put all vectors into the columns of a matrix, the number of pivots is 2.
- ○ The three vectors are linearly dependent.
- ○ There are exactly 2 linear independent vectors.
- ○ We may linearly combine x and y to generate z.
- ◉ If we put all vectors into the columns of a matrix, the matrix is nonsingular.

Correct

5. Which of the following statements is not true? **1 / 1 point**

- ○ The time complexity of Gaussian elimination is $O(n^3)$ for an nxn linear system.
- ◉ A linear system must be singular when a zero appears in a pivot position during the Gaussian elimination process.
- ○ If a matrix A is invertible, then A^{-1} must be unique.
- ○ A square matrix is nonsingular if and only if it is invertible.
- ○ If we have m n-dimensional vectors, they must be linearly dependent if $m > n$.

Correct